

Today's Agenda

1) Transformers

↳ Attention is All You Need

Seq 2 Seq

Self-Attention

Context

Not on Proximity.

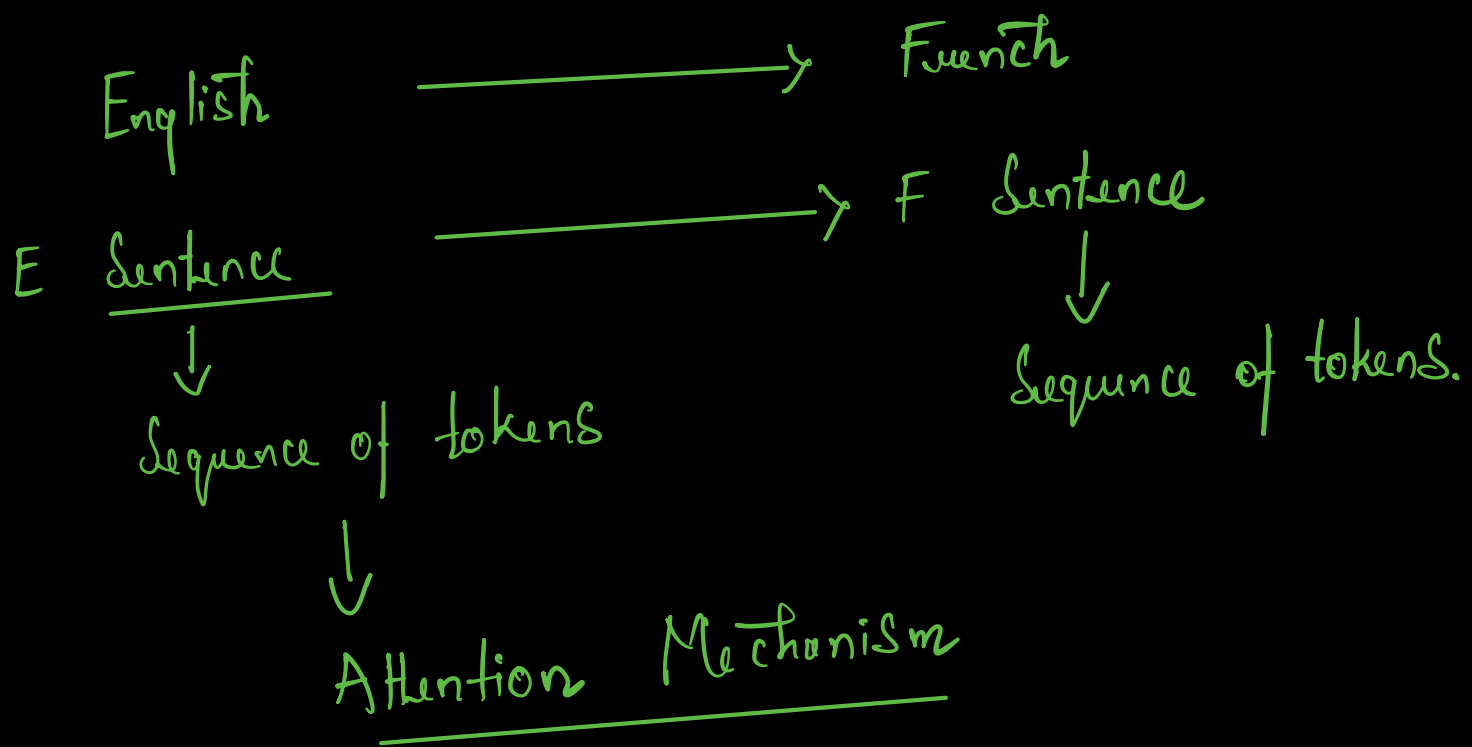
My name is Paul and I am going London

Sequence 2 Sequence

Encoder Decoder

RNN :- Many to Many

Machine Translation, Text Summarization, Text Generation

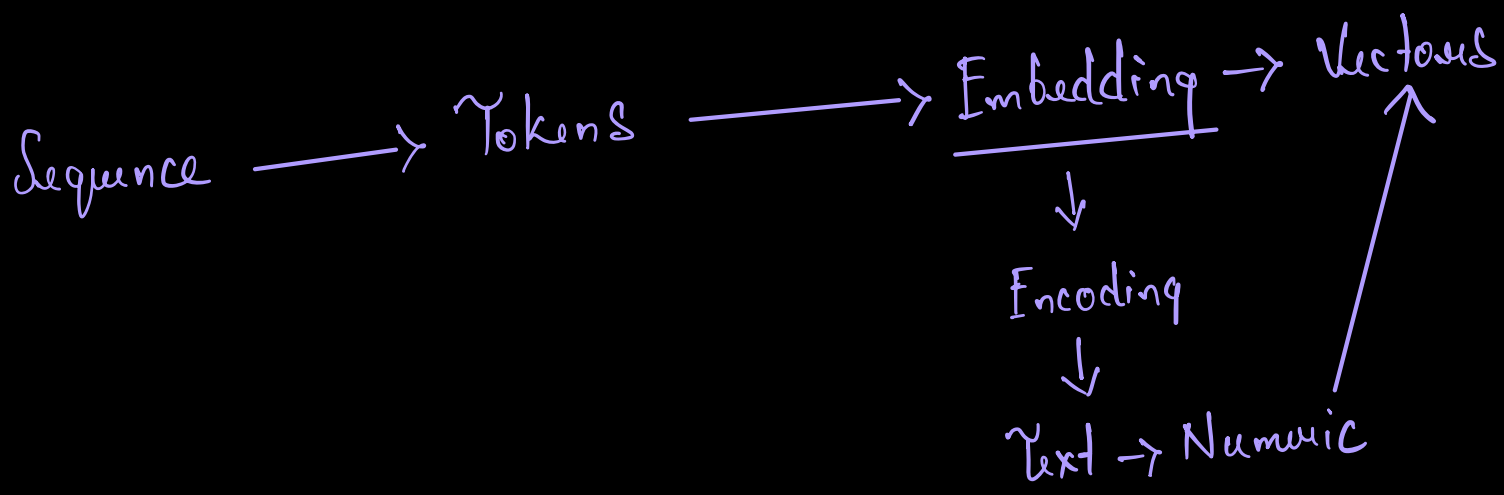


2022 LLM :- Large Language Models

Parameters :- 10 Billion

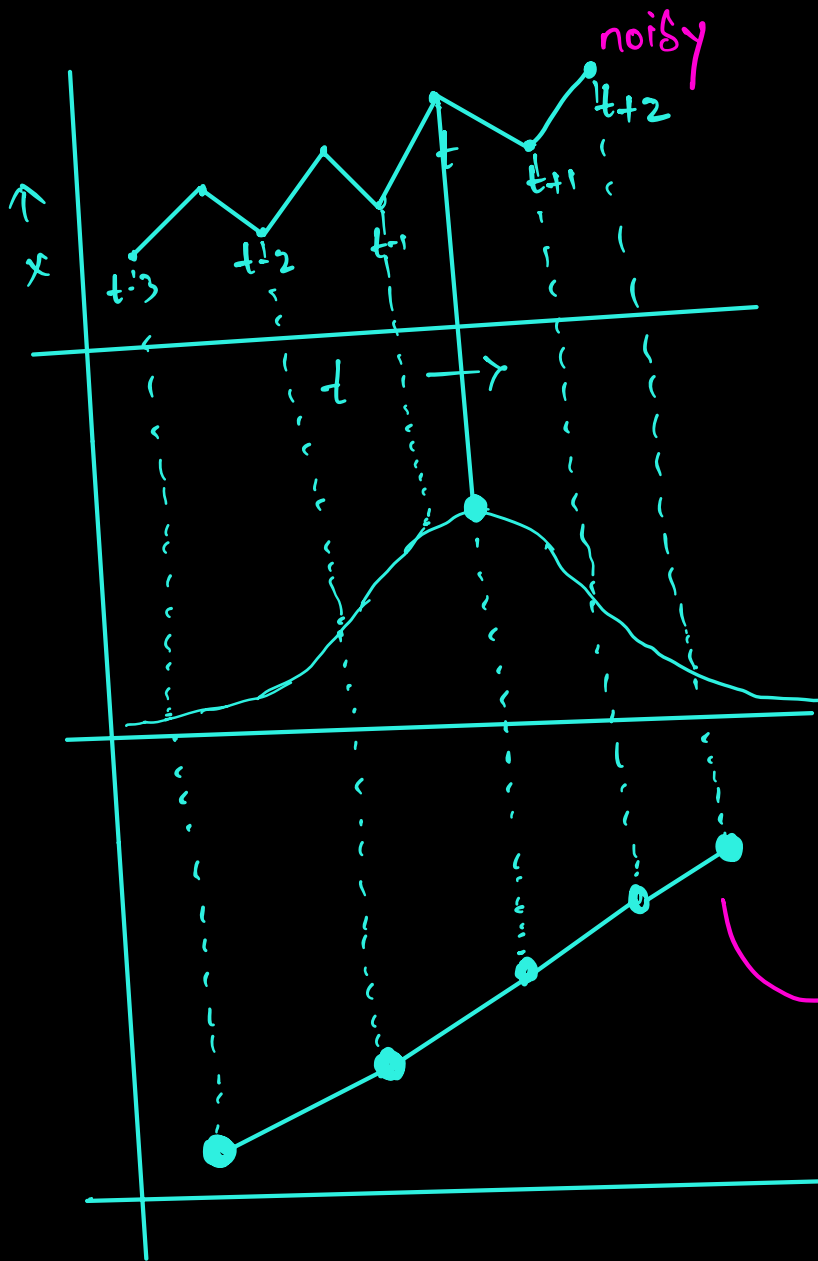
2018 SLM :- Small Language Models

500M $\sim$ 10 Billion



ATTENTION MECHANISMS

Time Series Data



EWA
Exponential Weighted Avg

Reweighing Scheme

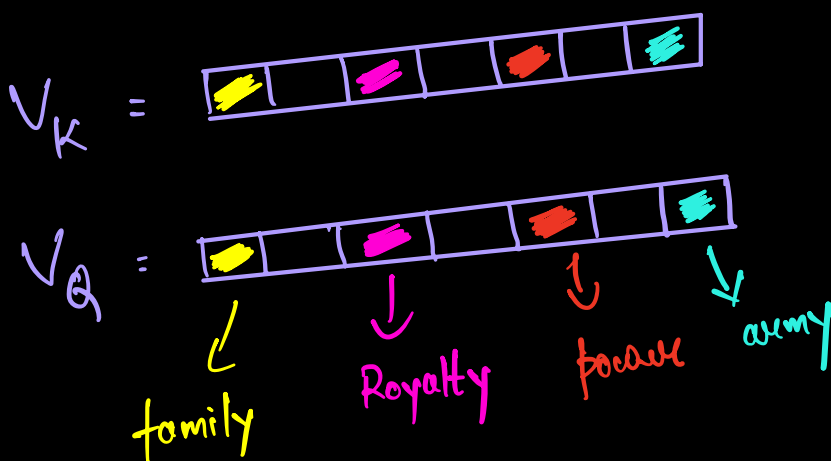
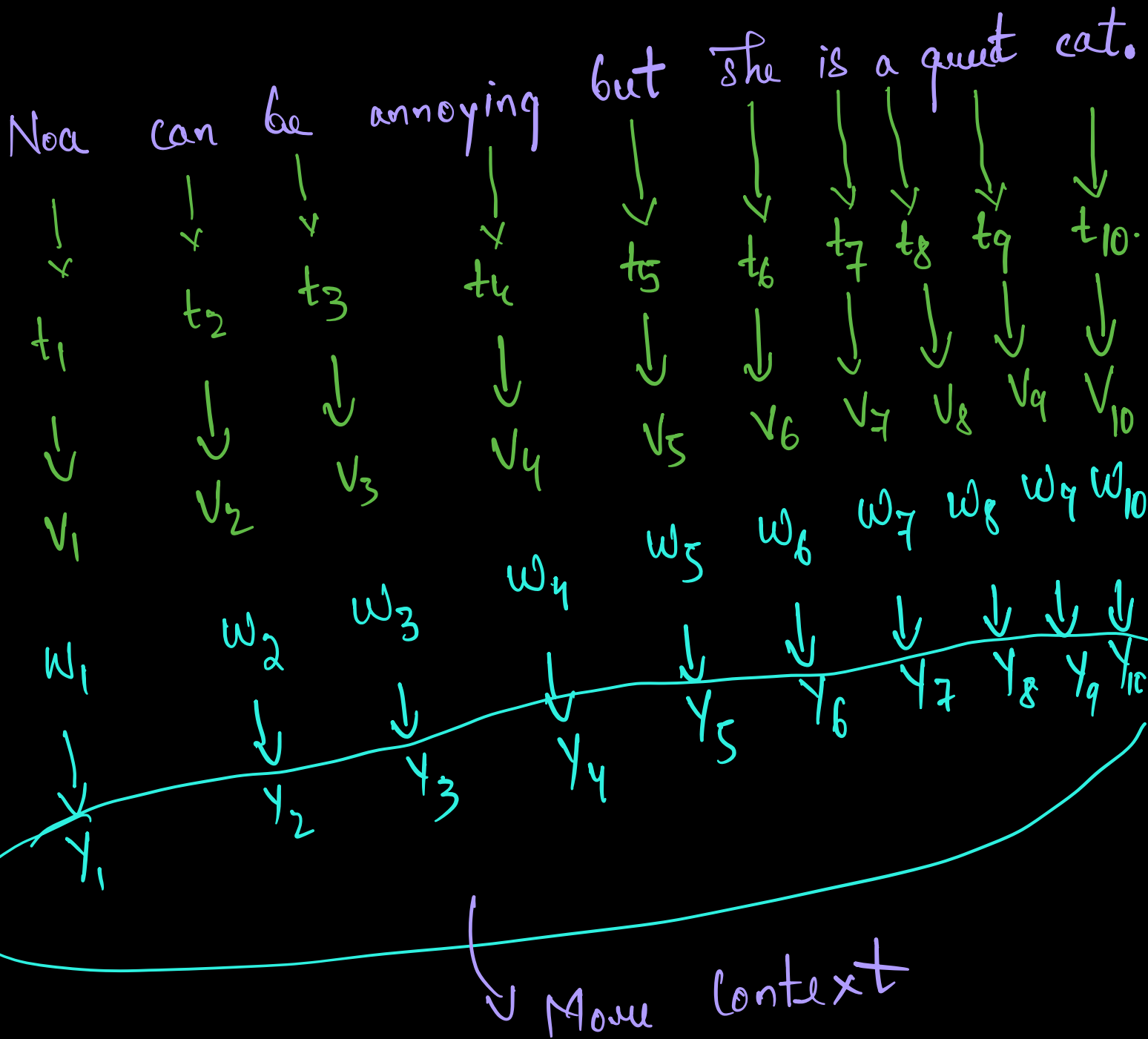
$$W_i = [w_0, w_1, w_2, w_3]$$

1) Filter data that is far away.

2) Amplify if something is close.

Useful data points gave context with each another.

Focus on the Reweighing Scheme.



Similar Words

Son
daughter
army
Kingdom
...

Reweighing Scheme / Plan

$$W_{kQ} = V_Q \cdot V_k$$

$$= V_k - V_Q$$

→ Explode

country
land

Dog, cat, computer
Swimming